## frq4

**1.** Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

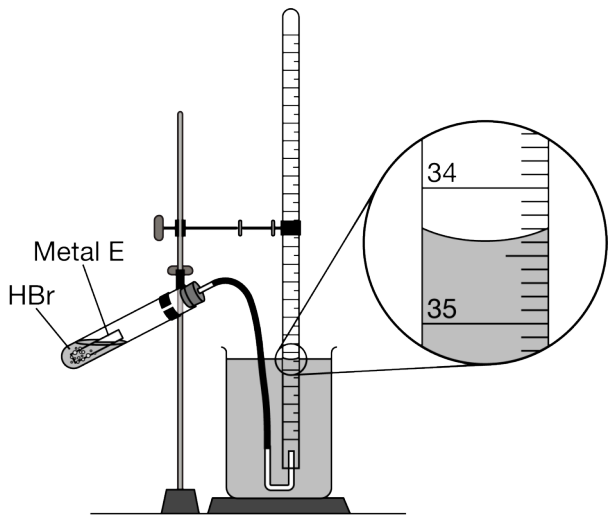
While investigating the chemical and physical properties of metallic element $\mathrm{E}$, a student conducts an experiment by combining $\mathrm{E}(s)$ and $\mathrm{HBr}(aq)$, which react according to the following equation.

$$2\ \mathrm{E}(s) + 2\ \mathrm{HBr}(aq) \rightarrow 2\ \mathrm{EBr}(aq) + \mathrm{H_2}(g)$$

(a) Write the net-ionic equation for the reaction.

(b) In the reaction, is $\mathrm{E}(s)$ being oxidized or reduced? Justify your answer in terms of oxidation numbers.

As shown in the following diagram, $10.00\ \mathrm{mL}$ samples of $\mathrm{HBr}(aq)$ are added to different quantities of $\mathrm{E}(s)$ in a closed test tube. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of $\mathrm{E}(s)$ react with $\mathrm{HBr}(aq)$, and the $\mathrm{H_2}(g)$ produced is collected in the gas-collecting tube.




(c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in $\mathrm{mL}$, of $\mathrm{H_2}(g)$ collected in trial 3?

(d) In the laboratory, the temperature is $295\ \mathrm{K}$ and the total pressure in the gas-collecting tube is $748.6\ \mathrm{mmHg}$. If the vapor pressure of water at $295\ \mathrm{K}$ is $19.8\ \mathrm{mmHg}$, determine the pressure of the $\mathrm{H_2}(g)$ in the gas-collecting tube.

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| Trial | Initial Moles of $E(s)$ | Initial $[HBr(aq)]$ $(M)$ | Volume of $H_2(g)$ Produced $(mL)$ |
|---|---|---|---|
| 1 | 0.00220 | 0.500 | 27.02 |
| 2 | 0.00240 | 0.500 | 29.40 |
| 3 | 0.00280 | 0.500 | ? |

(e) The data from all three trials are given above. A student claims that $HBr(aq)$ is the limiting reactant in trial 1. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element $E$ is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

| Element | Moles Present in the Compound |
|---|---|
| E | 0.0676 |
| S | 0.0338 |
| O | 0.102 |

(g) A different element, $L$, is located in the same row of the periodic table as element $E$, but $L$ has roughly half the atomic mass of $E$. Is the atomic radius of $L$ less than, equal to, or greater than the atomic radius of $E$? Justify your answer using principles of atomic structure.

(h) The ionic radii of elements $E$ and a different metallic element, $M$, are shown in the following table. Both elements form oxides, $E_2O$ and $MO$. If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of $MO$ be less than, equal to, or greater than the lattice energy of the oxide $E_2O$? Justify your answer in terms of Coulomb's law.

| Element | Ionic radius $(pm)$ |
|---|---|
| $E$ | 170 |
| $M$ | 110 |

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**2.** **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

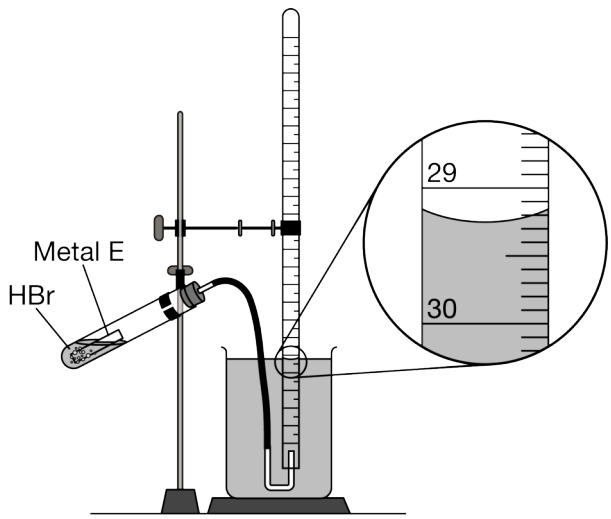
While investigating the chemical and physical properties of metallic element $\mathrm{E}$, a student conducts an experiment by combining $\mathrm{E}(s)$ and $\mathrm{HBr}(aq)$, which react according to the following equation.

$$2\ \mathrm{E}(s) + 6\ \mathrm{HBr}(aq) \rightarrow 2\ \mathrm{EBr_3}(aq) + 3\mathrm{H_2}(g)$$

(a) Write the net-ionic equation for the reaction.

(b) In the reaction, is $\mathrm{E}(s)$ being oxidized or reduced? Justify your answer in terms of oxidation numbers.

As shown in the following diagram, $10.00\ \mathrm{mL}$ samples of $\mathrm{HBr}(aq)$ are added to different quantities of $\mathrm{E}(s)$ in a closed test tube. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of $\mathrm{E}(s)$ react with $\mathrm{HBr}(aq)$, and the $\mathrm{H_2}(g)$ produced is collected in the gas-collecting tube.




(c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in $\mathrm{mL}$, of $\mathrm{H_2}(g)$ collected in trial 3?

(d) In the laboratory, the temperature is $295\ \mathrm{K}$ and the total pressure in the gas-collecting tube is $744.7\ \mathrm{mmHg}$. If the vapor pressure of water at $295\ \mathrm{K}$ is $19.8\ \mathrm{mmHg}$, determine the pressure of the $\mathrm{H_2}(g)$ in the gas-collecting tube.

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| Trial | Initial Moles of $E(s)$ | Initial $[HBr(aq)]$ $(M)$ | Volume of $H_2(g)$ Produced (mL) |
|---|---|---|---|
| 1 | $5.40 \times 10^{-4}$ | 0.400 | 19.95 |
| 2 | $6.90 \times 10^{-4}$ | 0.400 | 25.52 |
| 3 | $8.20 \times 10^{-4}$ | 0.400 | ? |

(e) The data from all three trials are given above. A student claims that $E(s)$ is the limiting reactant in trial 2. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element $E$ is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

| Element | Moles Present in the Compound |
|---|---|
| E | 0.0167 |
| I | 0.050 |
| O | 0.200 |

(g) A different element, $L$, is located in the same column of the periodic table as element $E$, but $L$ has roughly double the atomic mass of $E$. Is the atomic radius of $L$ less than, equal to, or greater than the atomic radius of $E$? Justify your answer using principles of atomic structure.

(h) The ionic radii of ions $X^{2-}$ and $Z^{2-}$ are shown in the following table. Element $E$ can bond with $X$ or $Z$ to form $E_2X_3$ or $E_2Z_3$, respectively. If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of $E_2X_3$ be less than, equal to, or greater than the lattice energy of $E_2Z_3$? Justify your answer in terms of Coulomb's law.

| Element | Ionic radius (pm) |
|---|---|
| $X^{2-}$ | 120 |
| $Z^{2-}$ | 180 |

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**3.** Answer the following questions using principles of chemical bonding and molecular structure.

(a) Consider the carbon dioxide molecule, $CO_2$ , and the carbonate ion, $CO_3^{2-}$.

(i) Draw the complete Lewis electron-dot structure for each species.

(ii) Account for the fact that the carbon-oxygen bond length in $CO_3^{2-}$ is greater than the carbon-oxygen bond length in $CO_2$.

(b) Consider the molecules $CF_4$, and $SF_4$ .

(i) Draw the complete Lewis electron-dot structure for each molecule.

(ii) In terms of molecular geometry, account for the fact that the $CF_4$ molecule is nonpolar, whereas the $SF_4$ molecule is polar.

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**4. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

An unknown gas, $\mathrm{D}$, reacts with fluorine gas to form the compound $\mathrm{DF_4}(g)$, as represented by the following equation.

$$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g)$$

A series of experimental trials were performed at $600^\circ\mathrm{C}$ to determine the rate law expression for the reaction. Data from the trials are shown below.

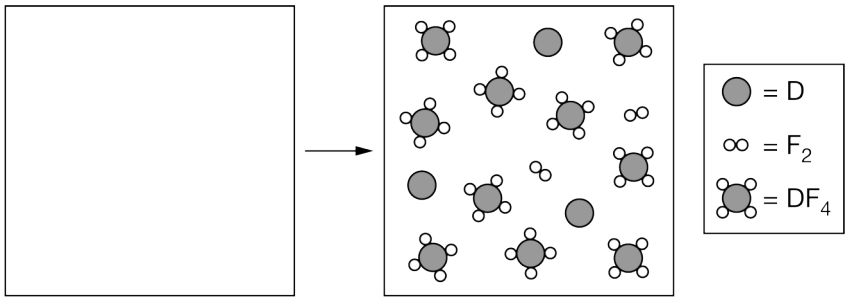
| | Initial $P_{\mathrm{D}}$ (torr) | Initial $P_{\mathrm{F_2}}$ (torr) | Initial Rate of Appearance of $\mathrm{DF_4}(g)$ (torr/min) |
|---|---|---|---|
| Trial 1 | 1500 | 1500 | $5.00 \times 10^{-3}$ |
| Trial 2 | 3000 | 1500 | $10.0 \times 10^{-3}$ |
| Trial 3 | 1500 | 3000 | $20.0 \times 10^{-3}$ |

(a) A student claims that the reaction is first order with respect to $\mathrm{F_2(g)}$. Do you agree or disagree? Justify your answer using the data above.

(b) What is the initial rate of disappearance of $\mathrm{F_2}(g)$ in trial 3? Justify your answer.

(c) A student claims that if element $\mathrm{D}$ is in group 18, then the molecular geometry of $\mathrm{DF_4}(g)$ is square planar. Do you agree or disagree with the student? Justify your answer in terms of $\mathrm{VSEPR}$ theory.

The particulate model shown below represents the moles of particles present in the container after $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to "Before Reaction" is intentionally left blank.)




Before Reaction After Reaction

**frq4**

(d) If the total pressure at the end of the reaction is $12\ \text{atm}$, what is the partial pressure of $\text{D}(g)$?

(e) Assuming that only $\text{D}$ and $\text{F}_2$ are present in the container before the reaction, determine the number of moles of $\text{D}$ and $\text{F}_2$ that were present in the initial reaction mixture.

$\text{D} =$ _____

$\text{F}_2 =$ _____

The equation and enthalpy change for the reaction between $\text{D}(g)$ and $\text{F}_2(g)$ are given below.

$$\text{D}(g) + 2\,\text{F}_2(g) \rightarrow \text{DF}_4(g) \quad \Delta H^\circ = -250\ \text{kJ/mol}_{rxn}$$

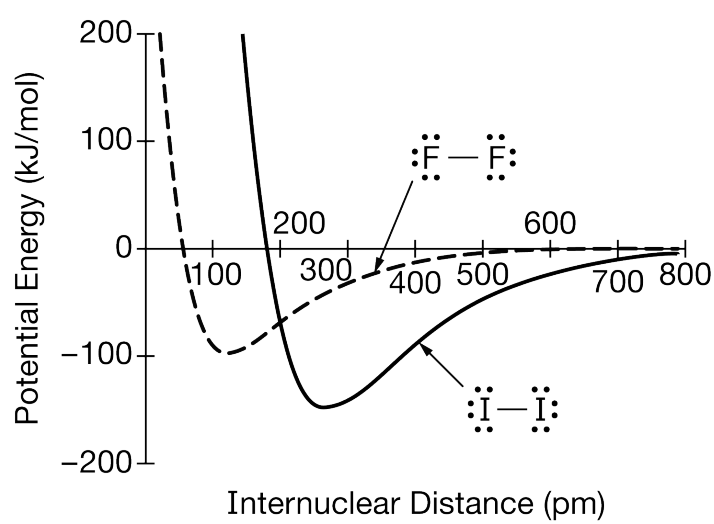
(f) If $0.50\ \text{mol}$ of $\text{DF}_4(g)$ is produced when equimolar amounts of $\text{D}(g)$ and $\text{F}_2(g)$ react, how much heat is released?

Fluorine can also react with another gas, $\text{E}(g)$, according to the following equation.

$$\text{E}(g) + 2\,\text{F}_2(g) \rightarrow \text{EF}_4(g) \quad \Delta\text{H}^\circ = -150\ \text{kJ/mol}_{rxn}$$

(g) The student claims that the $\text{D} - \text{F}$ bond energy is greater than the $\text{E} - \text{F}$ bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the diagram below, which shows the potential energy of two iodine atoms versus the distance between their nuclei. The student also finds that the $\text{I} - \text{I}$ bond energy is slightly less than the $\text{F} - \text{F}$ bond energy. The student <u>incorrectly</u> sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.




(h) Explain the error with the student's sketch for $\text{F}_2(g)$.

**frq4**

**5. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

An unknown gas, $\mathrm{D}$, reacts with fluorine gas to form the compound $\mathrm{DF_4}(g)$, as represented by the following equation.

$$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g)$$

A series of experimental trials were performed at $600^\circ\mathrm{C}$ to determine the rate law expression for the reaction. Data from the trials are shown below.

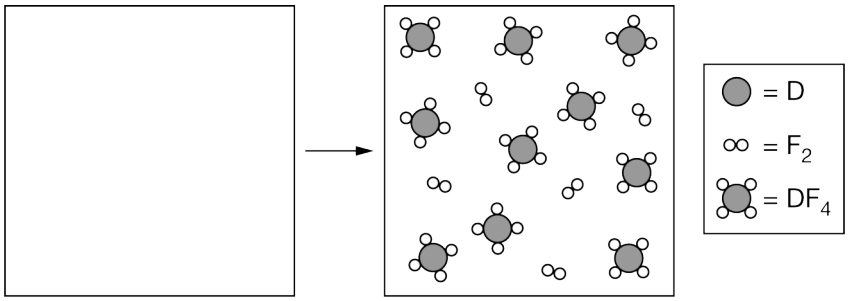
| | Initial $P_{\mathrm{D}}$ (torr) | Initial $P_{\mathrm{F_2}}$ (torr) | Initial Rate of Appearance of $\mathrm{DF_4}(g)$ (torr/min) |
|---|---|---|---|
| Trial 1 | 3000 | 3000 | $2.50 \times 10^{-2}$ |
| Trial 2 | 6000 | 3000 | $5.00 \times 10^{-2}$ |
| Trial 3 | 3000 | 6000 | $10.0 \times 10^{-2}$ |

(a) A student claims that the reaction is first order with respect to $\mathrm{D}(g)$. Do you agree or disagree? Justify your answer using the data above.

(b) What is the initial rate of disappearance of $\mathrm{F_2}(g)$ in trial 1? Justify your answer.

(c) A student claims that if element $\mathrm{D}$ is in group 18, then the molecular geometry of $\mathrm{DF_4}(g)$ is tetrahedral. Do you agree or disagree with the student? Justify your answer in terms of $\mathrm{VSEPR}$ theory.

The particulate model shown below represents the moles of particles present in the container after $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to "Before Reaction" is intentionally left blank.)



**frq4**

(d) If the total pressure at the end of the reaction is $12\ \text{atm}$, what is the partial pressure of $DF_4(g)$?

(e) Assuming that only $D$ and $F_2$ are present in the container before the reaction, determine the number of moles of $D$ and $F_2$ that were present in the initial reaction mixture.

$D =$ _____

$F_2 =$ _____

The equation and enthalpy change for the reaction between $D(g)$ and $F_2(g)$ are given below.

$$D(g) + 2\,F_2(g) \rightarrow DF_4(g) \quad \Delta H^\circ = -250\,\text{kJ/mol}_{rxn}$$

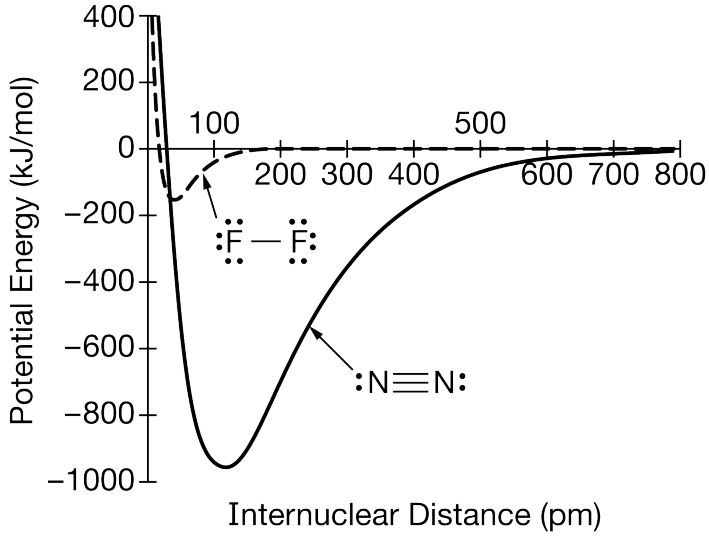
(f) How much heat is released when $4.0\ \text{mol}$ of $F_2(g)$ reacts with excess $D(g)$?

Fluorine can also react with another gas, $E(g)$, according to the following equation.

$$E(g) + 2\,F_2(g) \rightarrow EF_4(g) \quad \Delta H^\circ = -350\,\text{kJ/mol}_{rxn}$$

(g) The student claims that the $D - F$ bond energy is greater than the $E - F$ bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the diagram below, which shows the potential energy of two nitrogen atoms versus the distance between their nuclei. The student incorrectly sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.




(h) Explain the error with the student's sketch for $F_2(g)$.

**frq4**

**6.** **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

An unknown gas, $\mathrm{D}$, reacts with fluorine gas to form the compound $\mathrm{DF_4}(g)$, as represented by the following equation.

$$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g)$$

A series of experimental trials were performed at $600\,^{\circ}\mathrm{C}$ to determine the rate law expression for the reaction. Data from the trials are shown below.

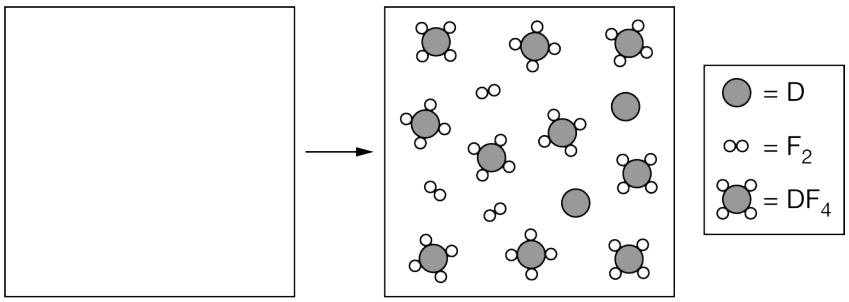
| | Initial $P_{\mathrm{D}}$ (torr) | Initial $P_{\mathrm{F_2}}$ (torr) | Initial Rate of Appearance of $\mathrm{DF_4}(g)$ (torr/min) |
|---|---|---|---|
| Trial 1 | 3000 | 3000 | $2.50 \times 10^{-2}$ |
| Trial 2 | 6000 | 3000 | $5.00 \times 10^{-2}$ |
| Trial 3 | 3000 | 6000 | $10.0 \times 10^{-2}$ |

(a) A student claims that the reaction is second order with respect to $\mathrm{F_2}(g)$. Do you agree or disagree? Justify your answer using the data above.

(b) What is the initial rate of disappearance of $\mathrm{D}(g)$ in trial 3? Justify your answer.

(c) A student claims that if element $\mathrm{D}$ is in group 16, then the molecular geometry of $\mathrm{DF_4}(g)$ is see-saw. Do you agree or disagree with the student? Justify your answer in terms of $\mathrm{VSEPR}$ theory.

The particulate model shown below represents the moles of particles present in the container after $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to "Before Reaction" is intentionally left blank.)




Before Reaction After Reaction

**frq4**

(d) If the total pressure at the end of the reaction is $12\ \text{atm}$, what is the partial pressure of $\text{D}(g)$?

(e) Assuming that only $\text{D}$ and $\text{F}_2$ are present in the container before the reaction, determine the number of moles of $\text{D}$ and $\text{F}_2$ that were present in the initial reaction mixture.

$\text{D} =$ _____

$\text{F}_2 =$ _____

The equation and enthalpy change for the reaction between $\text{D}(g)$ and $\text{F}_2(g)$ are given below.

$$\text{D}(g) + 2\,\text{F}_2(g) \rightarrow \text{DF}_4(g) \quad \Delta H^\circ = -250\ \text{kJ/mol}_{rxn}$$

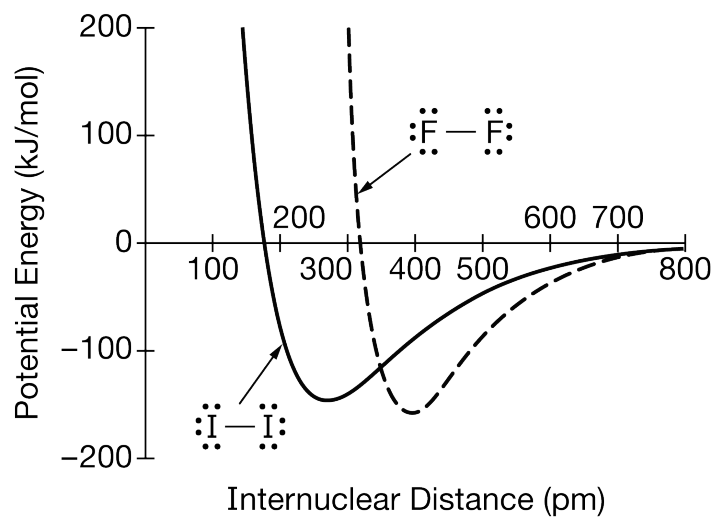
(f) How much heat is released when $0.75\ \text{mol}$ of $\text{D}(g)$ reacts with excess $\text{F}_2(g)$ ?

Fluorine can also react with another gas, $\text{E}(g)$, according to the following equation.

$$\text{E}(g) + 2\,\text{F}_2(g) \rightarrow \text{EF}_4(g) \quad \Delta H^\circ = -150\,\text{kJ/mol}_{rxn}$$

(g) The student claims that the $\text{D} - \text{F}$ bond energy is less than the $\text{E} - \text{F}$ bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the following diagram, which shows the potential energy of two iodine atoms versus the distance between their nuclei. The student also finds that the $\text{I} - \text{I}$ bond energy is slightly less than the $\text{F} - \text{F}$ bond energy. The student incorrectly sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.




(h) Explain the error with the student's sketch for $\text{F}_2(g)$.

**frq4**

**7. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

An unknown gas, $\mathrm{D}$, reacts with fluorine gas to form the compound $\mathrm{DF_4}(g)$, as represented by the following equation.

$$\mathrm{D}(g) + 2\,\mathrm{F_2}(g) \rightarrow \mathrm{DF_4}(g)$$

A series of experimental trials were performed at $600^\circ\mathrm{C}$ to determine the rate law expression for the reaction. Data from the trials are shown below.

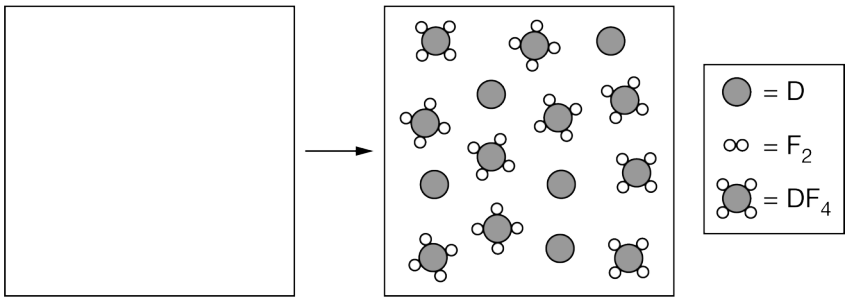
| | Initial $P_{\mathrm{D}}$ (torr) | Initial $P_{\mathrm{F_2}}$ (torr) | Initial Rate of Appearance of $\mathrm{DF_4}(g)$ (torr/min) |
|---|---|---|---|
| Trial 1 | 3000 | 3000 | $2.50 \times 10^{-2}$ |
| Trial 2 | 6000 | 3000 | $5.00 \times 10^{-2}$ |
| Trial 3 | 3000 | 6000 | $10.0 \times 10^{-2}$ |

(a) A student claims that the reaction is second order with respect to $\mathrm{D}(g)$. Do you agree or disagree? Justify your answer using the data above.

(b) What is the initial rate of disappearance of $\mathrm{F_2}(g)$ in trial 2? Justify your answer.

(c) A student claims that if element $\mathrm{D}$ is in group 16, then the molecular geometry of $\mathrm{DF_4}(g)$ is tetrahedral. Do you agree or disagree with the student? Justify your answer in terms of $\mathrm{VSEPR}$ theory.

The particulate model shown below represents the moles of particles present in the container after $\mathrm{D}(g)$ and $\mathrm{F_2}(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to "Before Reaction" is intentionally left blank.)




Before Reaction After Reaction

**frq4**

(d) If the total pressure at the end of the reaction is $12\ \text{atm}$, what is the partial pressure of $\text{D}(g)$?

(e) Assuming that only $\text{D}$ and $\text{F}_2$ are present in the container before the reaction, determine the number of moles of $\text{D}$ and $\text{F}_2$ that were present in the initial reaction mixture.

Math output error

The equation and enthalpy change for the reaction between $\text{D}(g)$ and $\text{F}_2(g)$ are given below.

$$\text{D}(g) + 2\,\text{F}_2(g) \rightarrow \text{DF}_4(g) \quad \Delta H^\circ = -250\,\text{kJ/mol}_{rxn}$$

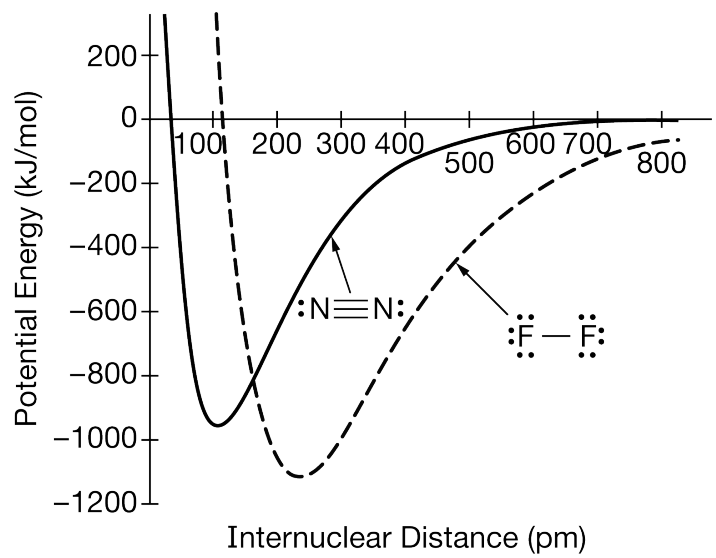
(f) How much heat is released when $0.50\ \text{mol}$ of $\text{F}_2(g)$ reacts with excess $\text{D}(g)$?

Fluorine can also react with another gas, $\text{E}(g)$, according to the following equation.

$$\text{E}(g) + 2\,\text{F}_2(g) \rightarrow \text{EF}_4(g) \quad \Delta H^\circ = -350\,\text{kJ/mol}_{rxn}$$

(g) The student claims that the $\text{D} - \text{F}$ bond energy is less than the $\text{E} - \text{F}$ bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the diagram below, which shows the potential energy of two nitrogen atoms versus the distance between their nuclei. The student incorrectly sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.




(h) Explain the error with the student's sketch for $\text{F}_2(g)$.

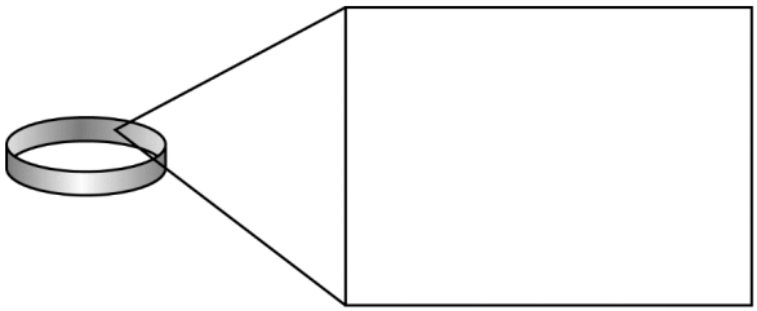
**frq4**

**8.** **Directions:** Examples and equations may be included in your responses where appropriate. For calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Pay attention to significant figures.

| Element | Atomic Radius (pm) |
|---|---|
| Au | 135 |
| Pd | 140 |

White gold is a common alloy of gold and palladium that is often used in jewelry. The atomic radii of the metals are given in the table above.

(a) A particular ring is made from an alloy that is 75 mole percent gold and 25 mole percent palladium. Using the box below, draw a particle-level diagram of the solid alloy consisting of 12 atoms with a representative proportion of atom types. Your diagram should clearly indicate whether the alloy is interstitial or substitutional. Use empty circles for gold and shaded circles for palladium.



White-gold jewelry is often coated with rhodium to modify the color and durability.

(b) A student hypothesizes that placing the ring in a 1.0 M solution of $Rh(NO_3)_3(aq)$ will result in a reaction in which Rh metal is deposited on the white-gold ring. Based on the information in the table below, calculate E° for the reactions that may occur between ions in the solution and atoms in the ring, and indicate whether or not the student's hypothesis is correct.

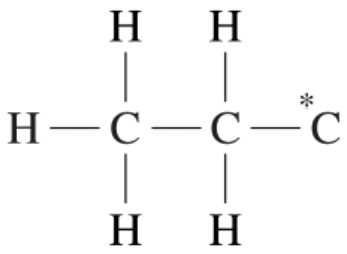
| Half-Reaction | E° (V) |
|---|---|
| $Au^{3+}(aq) + 3\,e^- \rightarrow Au(s)$ | 1.50 |
| $Pd^{2+}(aq) + 2\,e^- \rightarrow Pd(s)$ | 0.92 |
| $Rh^{3+}(aq) + 3\,e^- \rightarrow Rh(s)$ | 0.76 |

**frq4**

**9.** **Directions:** Examples and equations may be included in your responses where appropriate. For calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Pay attention to significant figures

Propanoic acid, C2H5COOH, is an organic acid that is a liquid at room temperature.

(a) An incomplete Lewis diagram for the propanoic acid molecule is provided in the box below. Complete the diagram, showing how the remaining atoms in the molecule are arranged around the carbon atom marked with an asterisk (*). Your structure should minimize formal charge and include any lone pairs of electrons.



(b) Identify the hybridization of the carbon atom marked with the asterisk

(c) Propanoic acid has a lower boiling point than butanoic acid, $C_3H_7COOH$.

i. Identify all the types of intermolecular forces present among the molecules in propanoic acid.

ii. Which of the types of intermolecular forces that you identified in part (c)(i) is most responsible for the difference in boiling points of the two acids?

## frq4

**10. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

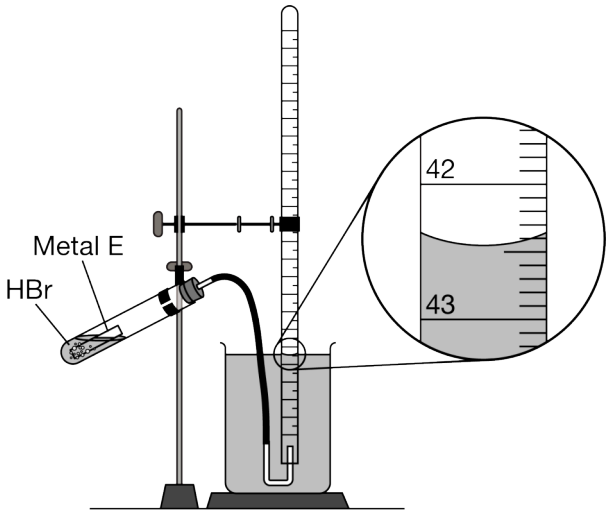
While investigating the chemical and physical properties of metallic element $\mathrm{E}$, a student conducts an experiment by combining $\mathrm{E}(s)$ and $\mathrm{HBr}(aq)$, which react according to the following equation.

$$\mathrm{E}(s) + 2\,\mathrm{HBr}(aq) \rightarrow \mathrm{EBr_2}(aq) + \mathrm{H_2}(g)$$

(a) Write the net-ionic equation for the reaction.

(b) In the reaction, is $\mathrm{E}(s)$ oxidized or reduced? Justify your answer in terms of oxidation numbers.

As shown in the following diagram, $10.00\,\mathrm{mL}$ samples of $\mathrm{HBr}(aq)$ of various concentrations are added to samples of $\mathrm{E}(s)$ in a closed test tube. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of $\mathrm{E}(s)$ react with $\mathrm{HBr}(aq)$, and the $\mathrm{H_2}(g)$ produced is collected in the gas-collecting tube.




(c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in $\mathrm{mL}$, of $\mathrm{H_2}(g)$ collected in trial 3?

(d) In the laboratory, the temperature is $294\,\mathrm{K}$ and the total pressure in the gas-collecting tube is $754.7\,\mathrm{mmHg}$. If the vapor pressure of water at $294\,\mathrm{K}$ is $18.6\,\mathrm{mmHg}$, determine the pressure of the $\mathrm{H_2}(g)$ in the gas-collecting tube.

**frq4**

| Trial | Initial Moles of $E(s)$ | Initial $[HBr(aq)]$ $(M)$ | Volume of $H_2(g)$ Produced (mL) |
|---|---|---|---|
| 1 | 0.00150 | 0.200 | 24.52 |
| 2 | 0.00150 | 0.250 | 30.95 |
| 3 | 0.00150 | 0.350 | ? |

(e) The data from all three trials are given above. A student claims that $E(s)$ is the limiting reactant in trial 1. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element $E$ is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

| Element | Moles Present in the Compound |
|---|---|
| E | 0.0652 |
| C | 0.130 |
| O | 0.260 |

(g) A different element, $L$, is located in the same row of the periodic table as element $E$, but $L$ has roughly double the atomic mass of $E$. Is the atomic radius of $L$ less than, equal to, or greater than the atomic radius of $E$? Justify your answer using principles of atomic structure.

(h) The ionic radii of elements $E$ and a different metallic element, $M$, are shown in the following table. Both elements form oxides, $EO$ and $M_2O$. If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of $M_2O$ be less than, equal to, or greater than the lattice energy of the oxide $EO$? Justify your answer in terms of Coulomb's law.

| Element | Ionic radius (pm) |
|---|---|
| E | 110 |
| M | 150 |

**frq4**

**11. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

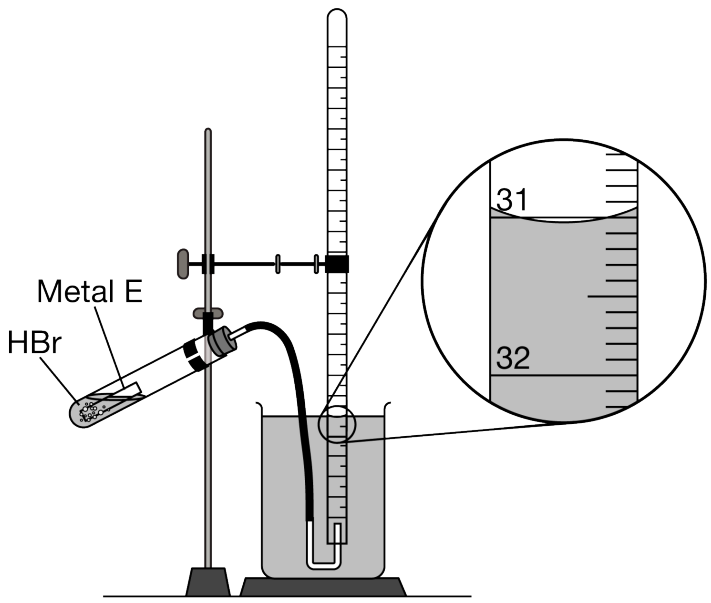
While investigating the chemical and physical properties of metallic element $\mathrm{E}$, a student conducts an experiment by combining $\mathrm{E}(s)$ and $\mathrm{HBr}(aq)$, which react according to the following equation.

$$\mathrm{E}(s) + 4\,\mathrm{HBr}(aq) \rightarrow \mathrm{EBr_4}(aq) + 2\,\mathrm{H_2}(g)$$

(a) Write the net-ionic equation for the reaction.

(b) In the reaction, is $\mathrm{H}$ being oxidized or reduced? Justify your answer using oxidation numbers.

As shown in the following diagram, in a closed test tube, $10.00\,\mathrm{mL}$ samples of $\mathrm{HBr}(aq)$ of various concentrations are added to samples of $\mathrm{E}(s)$. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of $\mathrm{E}(s)$ react with $\mathrm{HBr}(aq)$, and the $\mathrm{H_2}(g)$ produced is collected in the gas-collecting tube.




(c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in $\mathrm{mL}$, of $\mathrm{H_2}(g)$ collected in trial 3?

(d) In the laboratory, the temperature is $295\,\mathrm{K}$ and the total pressure in the gas-collecting tube is $741.2\,\mathrm{mmHg}$. If the vapor pressure of water at $295\,\mathrm{K}$ is $19.8\,\mathrm{mmHg}$, determine the pressure of the $\mathrm{H_2}(g)$ in the gas-collecting tube.

**frq4**

| Trial | Initial Moles of $E(s)$ | Initial $[HBr(aq)]$ $(M)$ | Volume of $H_2(g)$ Produced (mL) |
|---|---|---|---|
| 1 | $7.50 \times 10^{-4}$ | 0.150 | 18.60 |
| 2 | $7.50 \times 10^{-4}$ | 0.200 | 24.81 |
| 3 | $7.50 \times 10^{-4}$ | 0.250 | ? |

(e) The data from all three trials are given above. A student claims that $E(s)$ is the limiting reactant in trial 2. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element $E$ is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

| Element | Moles Present in the Compound |
|---|---|
| E | 0.0339 |
| C | 0.136 |
| O | 0.272 |

(g) A different element, $L$, is located in the same column of the periodic table as element $E$, but $L$ has roughly half the atomic mass of $E$. Is the atomic radius of $L$ less than, equal to, or greater than the atomic radius of $E$? Justify your answer using principles of atomic structure.

(h) The ionic radii of ions $X^-$ and $Z^-$ are shown in the following table. Element $E$ can bond with $X$ or $Z$ to form $EX_4$ or $EZ_4$, respectively. If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of $EX_4$ be less than, equal to, or greater than the lattice energy of $EZ_4$? Justify your answer in terms of Coulomb's law.

| Element | Ionic radius (pm) |
|---|---|
| $X^-$ | 170 |
| $Z^-$ | 120 |

**frq4**

**12.**

| Nonmetal | C | N | O | Ne | Si | P | S | Ar |
|---|---|---|---|---|---|---|---|---|
| Formula of Compound | $CF_4$ | $NF_3$ | $OF_2$ | No compound | $SiF_4$ | $PF_3$ | $SF_2$ | No compound |

Some binary compounds that form between fluorine and various nonmetals are listed in the table above. A student examines the data in the table and poses the following hypothesis: the number of F atoms that will bond to a nonmetal is always equal to 8 minus the number of valence electrons in the nonmetal atom.

a. Based on the student's hypothesis, what should be the formula of the compound that forms between chlorine and fluorine?

b. In an attempt to verify the hypothesis, the student researches the fluoride compounds of the other halogens and finds the formula $ClF_3$. In the box below, draw a complete Lewis electron-dot diagram for a molecule of $ClF_3$.

c. Two possible geometric shapes for the $ClF_3$ molecule are trigonal planar and T-shaped. The student does some research and learns that the molecule has a dipole moment. Which of the two shapes is consistent with the fact that the $ClF_3$ molecule has a dipole moment? Justify your answer in terms of bond polarity and molecular structure.

In an attempt to resolve the existence of the $ClF_3$ molecule with the hypothesis stated above, the student researches the compounds that form between halogens and fluorine, and assembles the following list.

| Halogen | Formula(s) |
|---|---|
| F | $F_2$ |
| Cl | |
| Br | $BrF$, $BrF_3$, $BrF_5$ |
| I | $IF$, $IF_3$, $IF_5$, $IF_7$ |

d. Based on concepts of atomic structure and periodicity, propose a modification to the student's previous hypothesis to account for the compounds that form between halogens and fluorine.

**frq4**

**13.**

| Nonmetal | C | N | O | Ne | Si | P | S | Ar |
|---|---|---|---|---|---|---|---|---|
| Formula of Compound | $CF_4$ | $NF_3$ | $OF_2$ | No compound | $SiF_4$ | $PF_3$ | $SF_2$ | No compound |

Some binary compounds that form between fluorine and various nonmetals are listed in the table above. A student examines the data in the table and poses the following hypothesis: the number of F atoms that will bond to a nonmetal is always equal to 8 minus the number of valence electrons in the nonmetal atom.

a. Based on the student's hypothesis, what should be the formula of the compound that forms between chlorine and fluorine?

b. In an attempt to verify the hypothesis, the student researches the fluoride compounds of the other halogens and finds the formula $ClF_3$. In the box below, draw a complete Lewis electron-dot diagram for a molecule
of $ClF_3$.

c. Two possible geometric shapes for the $ClF_3$ molecule are trigonal planar and T-shaped. The student does some research and learns that the molecule has a dipole moment. Which of the two shapes is consistent with the fact that the $ClF_3$ molecule has a dipole moment? Justify your answer in terms of bond polarity and molecular structure.

In an attempt to resolve the existence of the $ClF_3$ molecule with the hypothesis stated above, the student researches the compounds that form between halogens and fluorine, and assembles the following list.

| Halogen | Formula(s) |
|---|---|
| F | $F_2$ |
| Cl | |
| Br | $BrF$, $BrF_3$, $BrF_5$ |
| I | $IF$, $IF_3$, $IF_5$, $IF_7$ |

d. Based on concepts of atomic structure and periodicity, propose a modification to the student's previous hypothesis to account for the compounds that form between halogens and fluorine.

**frq4**

**14.** For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to $\mathrm{HBr}(l)$ and $\mathrm{HF}(l)$.

(a) In the following table, list all of the types of intermolecular forces present in pure samples of $\mathrm{HBr}(l)$ and $\mathrm{HF}(l)$.

| Liquid | $\mathrm{HBr}(l)$ | $\mathrm{HF}(l)$ |
|---|---|---|
| Intermolecular forces present | | |

(b) The enthalpy of vaporization, $\Delta H^{\circ}_{vap}$, for each liquid is provided in the following table.

| Liquid | $\mathrm{HBr}(l)$ | $\mathrm{HF}(l)$ |
|---|---|---|
| $\Delta H^{\circ}_{vap}$ | $17.3\ \mathrm{kJ/mol}$ | $25.2\ \mathrm{kJ/mol}$ |

(i) Based on the types and relative strengths of intermolecular forces, explain why $\Delta H^{\circ}_{vap}$ of $\mathrm{HF}(l)$ is greater than that of $\mathrm{HBr}(l)$.

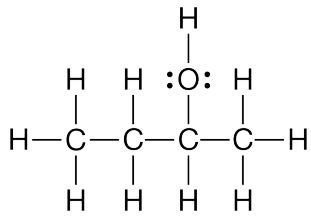
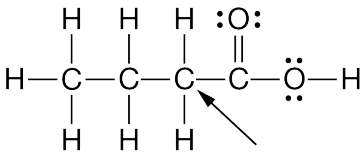
(ii) Calculate the amount of thermal energy, in $\mathrm{kJ}$, required to vaporize $6.85\ \mathrm{g}$ of $\mathrm{HF}(l)$.

(c) Based on the arrangement of electrons in the $\mathrm{Br}$ and $\mathrm{F}$ atoms, explain why the bond length in an $\mathrm{HBr}$ molecule is greater than that in an $\mathrm{HF}$ molecule.

**frq4**

**15. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, $\mathrm{L}$ and $\mathrm{X}$, are shown in the following table.

| Compound | L | X |
|---|---|---|
| Diagram |  |  |
| Boiling Point | 371 K | 437 K |

(a) Identify the hybridization of the valence orbitals of the carbon atom in compound $\mathrm{X}$ that is indicated by the arrow in the diagram.

(b) The $\mathrm{C-O}$ bond in compound $\mathrm{L}$ is longer than any $\mathrm{C-H}$ bond in compound $\mathrm{X}$. Explain the reason for this difference using principles of atomic structure.

(c) For each compound, list all intermolecular forces present.

Compound $\mathrm{L}$ –
Compound $\mathrm{X}$ –

(d) Explain why the boiling point of compound $\mathrm{X}$ is higher than that of compound $\mathrm{L}$ based on the types and relative strengths of the intermolecular forces present in each substance.

(e) A $0.10\ \mathrm{mol}$ sample of compound $\mathrm{L}$ is placed in an evacuated, rigid container at $400.\ \mathrm{K}$ and $0.25\ \mathrm{atm}$. When more moles of compound $\mathrm{L}$ are added, the pressure increases to $0.75\ \mathrm{atm}$ at $400.\ \mathrm{K}$. Determine the number of moles of compound $\mathrm{L}$ now present in the container.

Another carbon-containing compound is $\mathrm{C_2H_6O}$, which reacts with $\mathrm{H_2CrO_4}$, as represented by the following equation.

$$\mathrm{C_2H_6O + H_2CrO_4 \rightarrow C_2H_4O + H_2CrO_3 + H_2O}$$

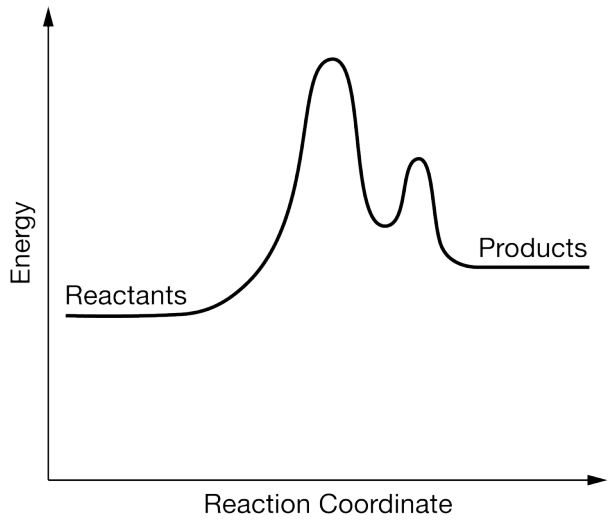
(f) The student claims that $\mathrm{C}$ is oxidized when $\mathrm{C_2H_6O}$ is converted to $\mathrm{C_2H_4O}$. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

**frq4**

The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

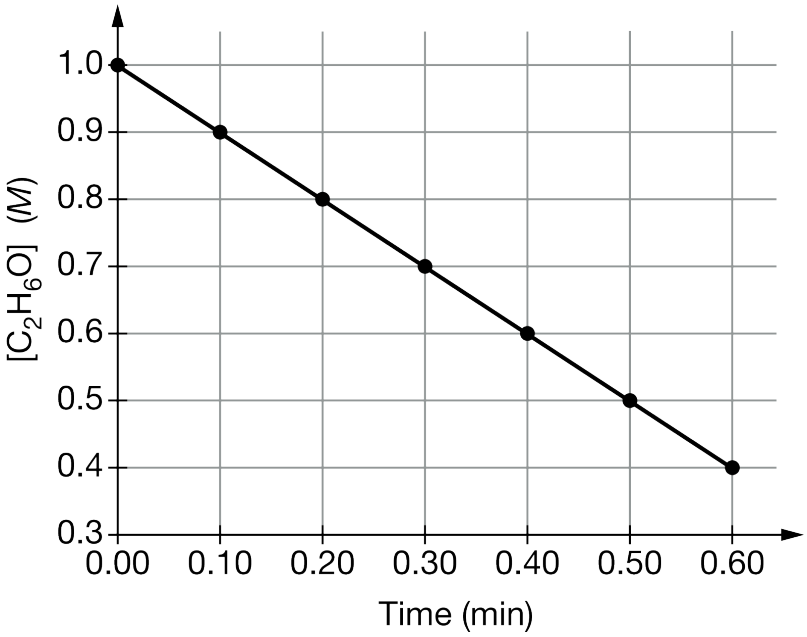
| | | |
|---|---|---|
| Step 1 | $C_2H_6O + H_2CrO_4 \rightleftarrows C_2H_5CrO_4H + H_2O$ | *fast* |
| Step 2 | $C_2H_5CrO_4H \rightarrow C_2H_4O + H_2CrO_3$ | *slow* |

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.




(h) The reaction is run in an excess of $H_2CrO_4$ such that $[H_2CrO_4]$ remains constant. The student claims that under these conditions, the reaction is zeroth-order with respect to $C_2H_6O$, and that the rate constant of the reaction is $1.0\ M/\text{min}$. Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.

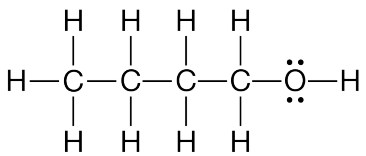
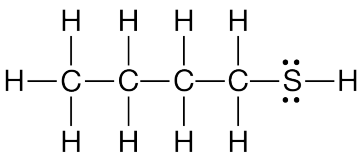
**frq4**


1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
[C2H6O] (M)
0.00
0.10
0.20
0.30
0.40
0.50
0.60
Time (min)

**frq4**

**16. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, $\mathrm{L}$ and $\mathrm{Z}$, are shown in the following table.

| Compound | L | Z |
|---|---|---|
| Diagram |  |  |
| Boiling Point | 391 K | 371 K |

(a) Based on $\mathrm{VSEPR}$ theory, what is the approximate value of the $\mathrm{C-O-H}$ bond angle in compound $\mathrm{L}$?

(b) The $\mathrm{C-O}$ bond in compound $\mathrm{L}$ is shorter than the $\mathrm{C-S}$ bond in compound $\mathrm{Z}$. Explain the reason for this difference using principles of atomic structure.

(c) For each compound, list all intermolecular forces present.

Compound $\mathrm{L}$ –
Compound $\mathrm{Z}$ –

(d) Explain why the boiling point of compound $\mathrm{L}$ is higher than that of compound $\mathrm{Z}$ based on the types and relative strengths of the intermolecular forces present in each substance.

(e) A gaseous sample of compound $\mathrm{Z}$ is placed in an evacuated, rigid container at a temperature of $400.\ \mathrm{K}$ and a pressure of $0.50\ \mathrm{atm}$. Determine the final pressure when the temperature of the sample is increased to $600.\ \mathrm{K}$.

Another carbon-containing compound is $\mathrm{C_2H_6O}$, which reacts with $\mathrm{H_2CrO_4}$, as represented by the following equation.

$$\mathrm{C_2H_6O + H_2CrO_4 \rightarrow C_2H_4O + H_2CrO_3 + H_2O}$$

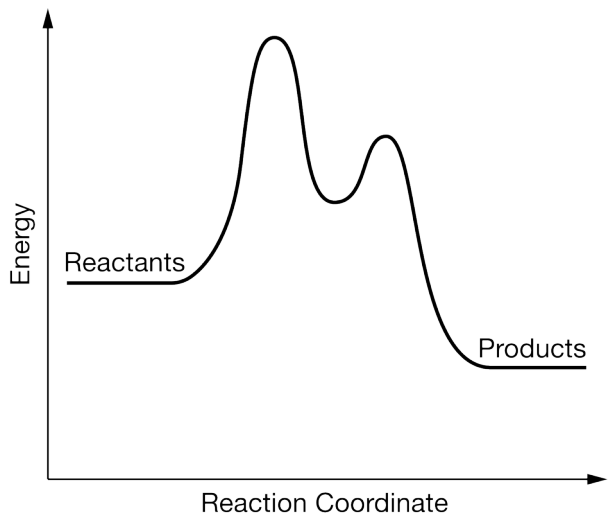
(f) The student claims that $\mathrm{O}$ is oxidized when $\mathrm{C_2H_6O}$ is converted to $\mathrm{C_2H_4O}$. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

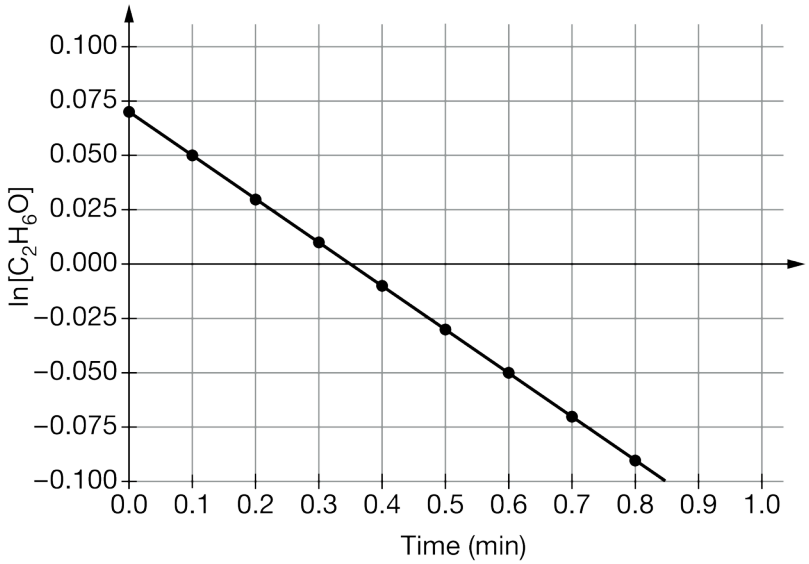
**frq4**

| | | |
|---|---|---|
| Step 1 | $C_2H_6O + H_2CrO_4 \rightleftarrows C_2H_5CrO_4H + H_2O$ | *fast* |
| Step 2 | $C_2H_5CrO_4H \rightarrow C_2H_4O + H_2CrO_3$ | *slow* |

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



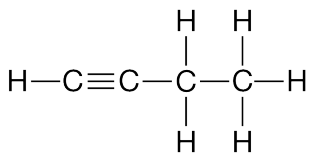
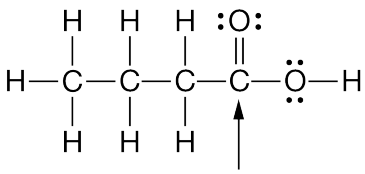

(h) The reaction is run in an excess of $H_2CrO_4$ such that $[H_2CrO_4]$ remains constant. The student claims that under these conditions, the reaction is first-order with respect to $C_2H_6O$, and that the rate constant of the reaction is $0.50\ \text{min}^{-1}$. Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.



**frq4**

**17. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, $\mathrm{Q}$ and $\mathrm{X}$, are shown in the following table.

| Compound | Q | X |
|---|---|---|
| Diagram |  |  |
| Boiling Point | 281 K | 437 K |

(a) Identify the hybridization of the valence orbitals of the carbon atom in compound $\mathrm{X}$ that is indicated by the arrow in the diagram.

(b) The $\mathrm{C-H}$ bonds in compound $\mathrm{Q}$ are shorter than the $\mathrm{C-C}$ bonds in compound $\mathrm{X}$. Explain the reason for this difference using principles of atomic structure.

(c) For each compound, list all intermolecular forces present.

Compound $\mathrm{Q}$ –
Compound $\mathrm{X}$ –

(d) Explain why the boiling point of compound $\mathrm{X}$ is higher than that of compound $\mathrm{Q}$ based on the types and relative strengths of the intermolecular forces present in each substance.

(e) A gaseous sample of compound $\mathrm{Q}$ is placed in an evacuated cylinder with a movable piston that can change the volume of the cylinder. The initial volume and temperature of the gas are $10.0\ \mathrm{L}$ and $400.\ \mathrm{K}$. If the temperature of the gas is raised to $500.\ \mathrm{K}$ as its pressure remains constant, what is the final volume of the gas?

Another carbon-containing compound is $\mathrm{C_2H_6O}$, which reacts with $\mathrm{H_2CrO_4}$, as represented by the following equation.

$$\mathrm{C_2H_6O + H_2CrO_4 \rightarrow C_2H_4O + H_2CrO_3 + H_2O}$$

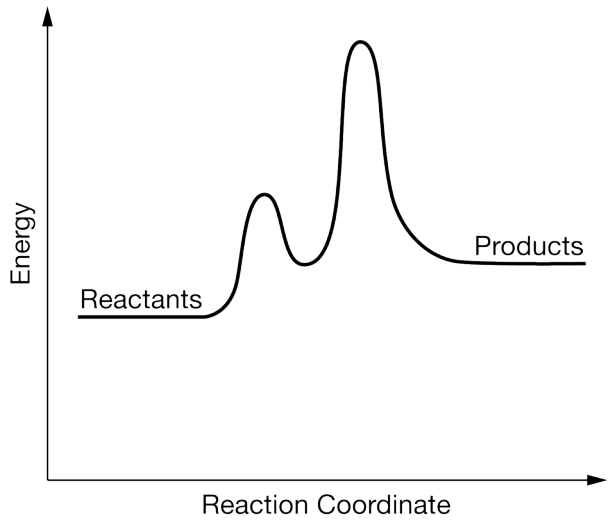
(f) The student claims that $\mathrm{Cr}$ is reduced when $\mathrm{H_2CrO_4}$ is converted to $\mathrm{H_2CrO_3}$. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

## frq4

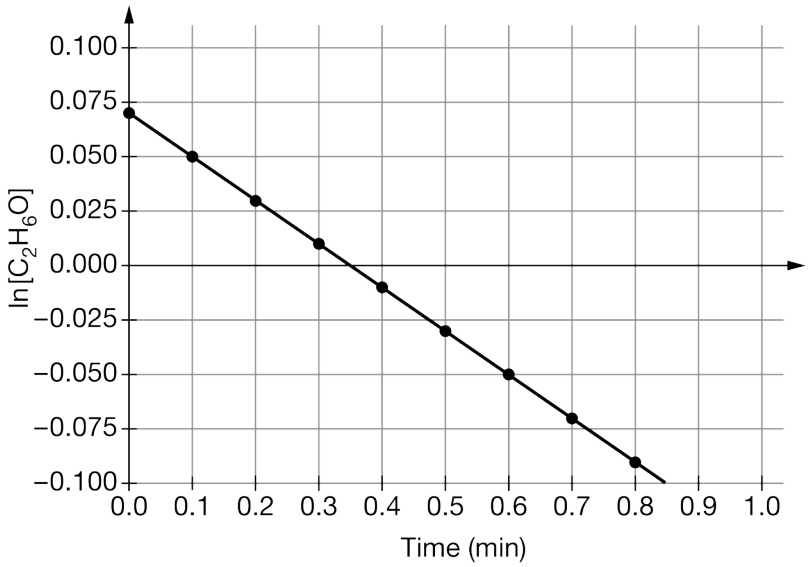
The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

| | | |
|---|---|---|
| Step 1 | $C_2H_6O + H_2CrO_4 \rightleftarrows C_2H_5CrO_4H + H_2O$ | *fast* |
| Step 2 | $C_2H_5CrO_4H \rightarrow C_2H_4O + H_2CrO_3$ | *slow* |

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



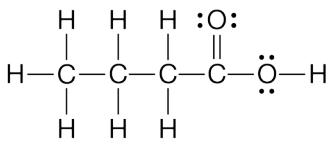
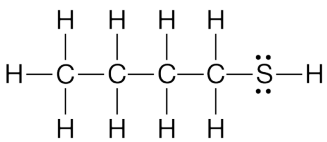

(h) The reaction is run in an excess of $H_2CrO_4$ such that $[H_2CrO_4]$ remains constant. The student claims that under these conditions, the reaction is second-order with respect to $C_2H_6O$, and that the rate constant of the reaction is $0.20\ M^{-1}\ \text{min}^{-1}$. Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.



**frq4**

**18. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, $\mathrm{X}$ and $\mathrm{Z}$, are shown in the following table.

| Compound | X | Z |
|---|---|---|
| Diagram |  |  |
| Boiling Point | 437 K | 371 K |

(a) Based on $\mathrm{VSEPR}$ theory, what is the approximate value of the $\mathrm{O-C-O}$ bond angle in compound $\mathrm{X}$?

b) The $\mathrm{O-H}$ bond in compound $\mathrm{X}$ is shorter than the $\mathrm{S-H}$ bond in compound $\mathrm{Z}$. Explain the reason for this difference using principles of atomic structure.

(c) For each compound, list all intermolecular forces present.

Compound $\mathrm{X}$ –
Compound $\mathrm{Z}$ –

(d) Explain why the boiling point of compound $\mathrm{X}$ is higher than that of compound $\mathrm{Z}$ based on the types and relative strengths of the intermolecular forces present in each substance.

(e) A gaseous sample of compound $\mathrm{Z}$ is placed in an evacuated, rigid container at a temperature of $800.\ \mathrm{K}$ and a pressure of $1.00\ \mathrm{atm}$. Determine the final pressure of the gas when the temperature is decreased to $600.\ \mathrm{K}$.

Another carbon-containing compound is $\mathrm{C_2H_6O}$, which reacts with $\mathrm{H_2CrO_4}$, as represented by the following equation.

$$\mathrm{C_2H_6O + H_2CrO_4 \rightarrow C_2H_4O + H_2CrO_3 + H_2O}$$

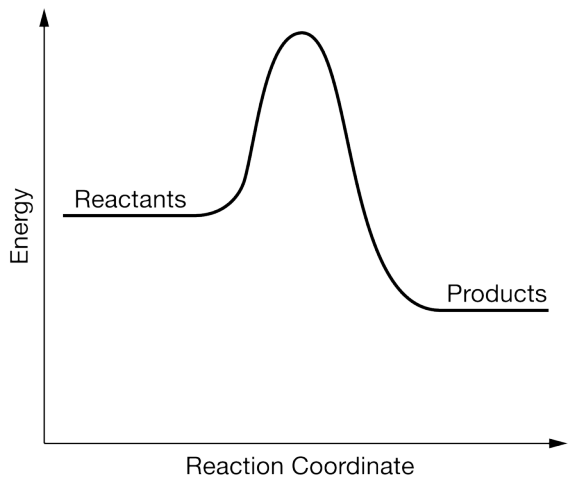
(f) The student claims that $\mathrm{C}$ is reduced when $\mathrm{C_2H_6O}$ is converted to $\mathrm{C_2H_4O}$. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

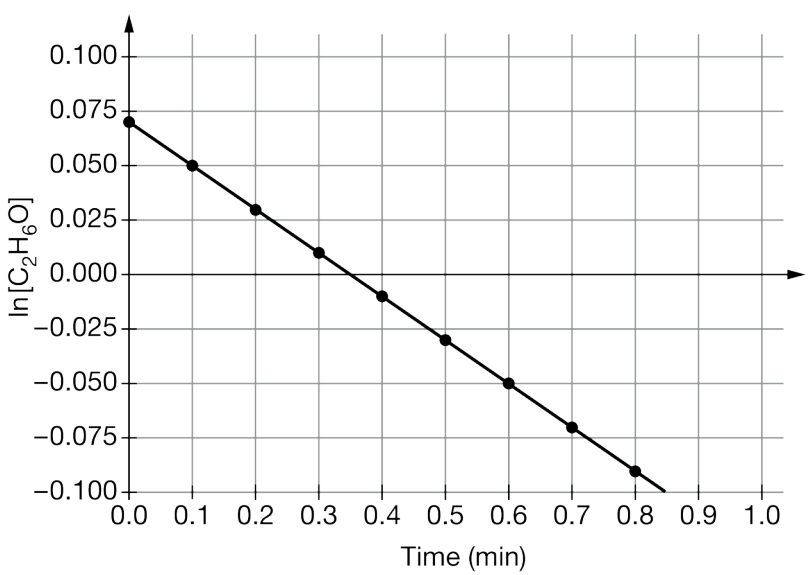
The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

**frq4**

| | | |
|---|---|---|
| Step 1 | $C_2H_6O + H_2CrO_4 \rightleftarrows C_2H_5CrO_4H + H_2O$ | *fast* |
| Step 2 | $C_2H_5CrO_4H \rightarrow C_2H_4O + H_2CrO_3$ | *slow* |

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.




(h) The reaction is run in an excess of $H_2CrO_4$ such that $[H_2CrO_4]$ remains constant. The student claims that under these conditions, the reaction is zeroth-order with respect to $C_2H_6O$, and that the rate constant of the reaction is $0.20\ M/\text{min}$. Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.